

PYTHON DAY 07

Control Flow

7.1 if Statement

Syntax

```
if condition:
    # code to execute
```

The **if** statement runs a block of code only when a condition is **True**. If the condition is False, Python skips the indented block.

Flow Chart (text form)

Condition → True: execute if block | False: skip if block

Example 1

```
age = 20

if age >= 18:
    print("Eligible to vote")
```

Output:

Eligible to vote

Example 2

```
marks = 35

if marks >= 40:
    print("Passed")

print("Program finished")
```

Output:

Program finished

7.2 if-else Statement

Syntax

```
if condition:
    # if block
else:
    # else block
```

The **if-else** statement chooses between two blocks. If the condition is True, the if block runs; otherwise, the else block runs. Only one block executes.

Flow Chart (text form)

Condition → True: if block | False: else block

Example

```
age = 16

if age >= 18:
```

```
        print("Eligible to vote")
    else:
        print("Not eligible to vote")
```

Output:

Not eligible to vote

7.3 elif Statement

Syntax

```
if condition1:
    # block 1
elif condition2:
    # block 2
else:
    # block 3
```

elif means **else if**. It is used to check multiple conditions. Python checks conditions from top to bottom and executes the first block whose condition is True. The **else** block runs only if all previous conditions are False.

Example

```
marks = 75

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 40:
    print("Grade C")
else:
    print("Fail")
```

Output:

Grade B

Quick Revision

if = one condition | if-else = two choices | elif = multiple conditions